

three station modules: Columbus, Kibo and Destiny. Another section, Benefits, gives information on health factors, Earth benefits and education scenarios across the globe.

You select a section and investigate a particular benefit. The media section has three divisions: Podcasts, Games and Videos. The Games section has a game that discusses different levels of gravity when a ball is tossed. The Podcasts section offers links to NASA ScienceCasts, while the Videos section provides links for science-related videos.

ISS works as a micro-gravity and space environment research laboratory between the orbital altitude of 330km and 435km. It travels at an average speed of 27,724km per hour. Crew members out there conduct experiments on biology, human biology, physics, astronomy, meteorology and other areas.

India to make the iris of the biggest telescope

India is set to contribute the iris of a 30m man-made eye, which will open atop the Hawaiian volcanic dome of Mauna Kea, sometime in 2023, to search for life beyond the solar system.

India has signed in as a full partner in the US\$ 1.4 billion thirty-metre telescope (TMT) project, which will be jointly developed by five nations, and is set to make sensors and actuators that will keep the huge mirror of the biggest telescope in place. Edge sensors and actuators are crucial components of the telescope as the huge mirror is not a single piece but comprises 492 hexagonal segments. Each segment is controlled by three activators and two edge sensors along each inter-segment gap to ensure accurate optical images.

B. Eswar Reddy, scientist, Indian Institute of Astrophysics (IIA), told the 102nd Indian Science Congress that India plays a cardinal role in setting-up the telescope that has a resolution 12 times better than that of Hubble Space Telescope.

The USA will make primary mirror segments, while China will make the tertiary mirror. Canada will have the dome put in place, and Japan the telescope structure. TMT is expected to strengthen the continuing search for intelligent life elsewhere, as it provides direct imaging of planetary systems. It will also help astronomers study planetary atmospheres, their origin and development.

Toshiba launches FlashAir



FlashAir, a secure digital high-capacity (SDHC) card by Toshiba, has embedded stand-alone wireless LAN functionality, which allows wireless transfer of data stored in these cards simultaneously to multiple devices, like PCs or smartphones, through multiple WLAN connections.

Design engineers can use FlashAir card to reduce development lead time, since it is not necessary to do HW design/development for the WLAN chipset or module portion of the system. This makes it easy to add wireless capability onto existing devices. This new card is fully compliant with existing SD card standards. Also, it uses very low power as wireless functionality is turned on only when necessary and remains redundant otherwise.

Gogoro launches zero-emission Smartscooter

A Taiwan based company, Gogoro, has come up with a new prototype of Smartscooter. The battery-swapping infrastructure is called Gogoro Energy Network. The company claims that this scooter gives out zero emissions.



Gogoro Smartscooter (Image courtesy: Gogoro)

Gogoro Smartscooter is based on a futuristic concept usually seen in sci-fi movies. It is full of amazing features and is powered by Gogoro G1 electric motor. The motor is fitted on a light-weight aerogramme monocoque chassis, which reduces the weight of the scooter. The scooter features more than 30 different kinds of sensors. It has jet-fighter landing-gear suspension in front and multi-link suspension on rear, which enables a stable and comfort ride.

The scooter can communicate with the Cloud, which is integrated with a dedicated mobile app. The built-in IQ system has the power of a PC, that lets the user synchronise the scooter with the app. The modular battery-swapping infrastructure can be deployed anywhere in the city. The mobile app can help in locating battery-vending machines called GoStations. Sensors are used to collect vital information to adjust top speed or range for riding patterns, which can be customised. The battery is replaceable, so when it is exhausted, it can be replaced with another at the GoStation within six seconds. All features on Gogoro Smartscooter are highly customisable.

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